

advantage of InfluxDB is its capacity to aggregate values in time buckets on-the-fly without any manual intervention.

InfluxDB can be accessed by software like Grafana, which is a powerful front-end tool providing visualisation features for time series data. InfluxDB has no external dependencies and SQL like queries are used for querying a data structure comprising measurements, series and points. Each point consists of varied key-value pairs called fieldset and timestamp. Values can be 64-bit integers, 64-bit floating points, strings and Booleans. Points are indexed by their time and tagset. InfluxDB stores data via HTTP, TCP and UDP.

Features

- Purely written in the Go programming language and facilitates compilation into a single binary with no external dependencies.
- High performance customised data store written especially for time series data. The TSM engine of InfluxDB allows efficient and high speed data storage and compression.
- Plugins support for other data ingestion protocols like Graphite, collectd, OpenTSDB.
- In-built Web front-end tool for database and user administration.
- Competent in merging multiple series together.

Official website: <https://www.influxdata.com/>

Latest version: 1.1.1

CrateDB

CrateDB is an open source distributed SQL database management system developed by Crate.io Inc., which fully integrates a searchable document-oriented data store. Christian Lutz, CEO of Crate.io, said, "When we founded Crate.io we set out to reinvent SQL for the machine data era. Today, 75 per cent of our customers use CrateDB for managing machine and IoT because of its easy usage, performance and versatility."

CrateDB makes machine data applications accessible to SQL developers; prior to this these were only possible using NoSQL solutions. CrateDB combines SQL with search versatility and ease of scalability of containers. It provides a good alternative to analytic data store tools like

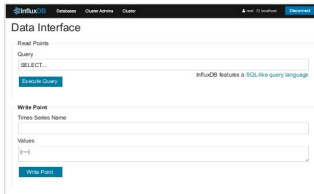


Figure 1: WebUI for InfluxDB

Splunk. The CrateDB platform includes the distributed SQL query engine for providing faster joins, aggregations and ad-hoc queries; SQL with integrated search for data and query versatility; and container architecture and automatic data sharding for simple scaling.

The main language used by CrateDB is SQL but it also makes use of the document-oriented approach of NoSQL style databases. It uses the SQL parser from Facebook Presto for its query and prediction analysis. It includes an in-built administration interface. The *Crate Shell CLI* allows users to put up interactive SQL queries.

Features

- **Highly scalable:** Updates to the database are easy and can be made by simply adding new machines to update the cluster; there is no need for any re-distribution of data in the cluster as it is done automatically by CrateDB.
- **Highly available:** CrateDB allows the database to be highly available if anything goes wrong, as it provides automated replication of data across the cluster; even hardware and software updates don't interrupt normal data operations. CrateDB has the capability of self-healing infected nodes.
- **Real-time data ingestion:** CrateDB delivers millisecond speed query performance even if writes are taking place, and removes locking overheads.
- **Supports various data:** CrateDB supports both relational as well as JSON-documents. And it also provides blob storage to store and retrieve videos, pictures or other unstructured files.
- It supports geospatial queries and dynamic schemas, making CrateDB fully flexible, which is very good for Agile based development and IoT database storage at the back-end.

Official website: <https://crate.io>

Latest version: 1.0.4

Riak time series database

Riak time series (TS) database from Basho is an open source, distributed NoSQL key-value stored optimised database for the Internet of Things (IoT). With this database, the user can associate a large number of data points with a specific point in time. It is based on masterless architecture, in which every node in the cluster is capable of serving read and write requests; the distributed database automatically co-locates, replicates and distributes the data across the cluster to achieve high performance and availability.

Riak TS database is highly optimised for data access requirements. It supports Apache Spark integration, which makes integration support possible for Spark streaming, dataframes and Spark SQL.

Riak TS can be installed directly on the data centre or public cloud. AWS Amazon Machine Images (AMI) are also available for this database to facilitate users to experience Riak TS in the AWS workspace.